# Deployment & Customer Handover Guide

Real-Time Deep Learning Road Lane Detection & Autonomous Driving Telemetry System

*Client Deliverable | Phase 7: DevOps Release | Date: 2026-10-23 | Status: Approved*

## 1. Executive Overview & Deliverable Scope

This Customer Handover Guide outlines the deployment, operation, and maintenance procedures for the Real-Time Deep Learning Road Lane Detection System. Engineered for enterprise autonomous vehicle fleet monitoring, simulation, and driver assistance, the deliverable provides a turnkey Streamlit dashboard powered by vanishing-point constrained line detection and temporal smoothing filters.

## 2. Deployment Options

Option A: Production Docker Container (Recommended)
• Execute: docker compose up --build -d
• Web URL: http://localhost:8501
• Advantages: Complete isolation, non-root execution, automated health monitoring.

Option B: Local Python Virtual Environment
• Requires: Python 3.10+ (x86_64), pip
• Setup: python -m venv venv && .\venv\Scripts\activate && pip install -r requirements.txt
• Launch: streamlit run src/app.py

Option C: Cloud Deployment (Streamlit Community Cloud / AWS ECS)
• Ready to deploy with included packages.txt (libgl1, libglib2.0-0) and requirements.txt.

## 3. End-User Operational Guide & Telemetry Cards

The interface translates computer vision mathematics into consumer-friendly displays with bracketed tooltips:

| Dashboard Indicator | State Displayed | Driver Action Required |
|---|---|---|
| Camera Tracking | 🟢 LOCKED (Both Lane Lines Tracked)<br>◯ SEARCHING (Scanning Road / No Lines Found) | LOCKED: Normal operation.<br>SEARCHING: Pay manual attention; lines obscured. |

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|---|---|---|
| Vehicle Lane Position | ☑ Centered (within 0.12m)<br>⚠ Drifting Right/Left (> 0.12m) | Steer slightly toward center if drifting badge appears. |
| Road Shape Ahead | 🛣 Straight Road<br>↪ Gentle Curve<br>⚠ Sharp Turn | Anticipate steering adjustments based on curve severity. |
| Driver Safety Alert | ✓ SAFE IN LANE (Normal)<br>🚨 DEPARTURE ALERT! (Flashing red) | Immediate correction required if Departure Alert triggers. |

## 4. Telemetry Export & Fleet Logging

At any point during image or video playback, operators can click '📥 Download Complete Telemetry Log (CSV)' to retrieve a standardized RFC 4180 CSV log file containing timestamped lateral offsets, road curvatures, and lane departure alert flags for fleet safety auditing.